TASK SPECIFICATION: P5-04 PHASE P5 — SMART PARKING & ALPR COMPUTER VISION GATE

Flex-Share Peer Parking Exchange & Rewards Engine

Git Branch: `feature/P5-04-flex-share` · Target Pull Request: feature/P5-04-flex-share

Core Objective & Technical Scope:
Intelligent spot-sharing marketplace allowing permit holders to temporarily release unused spots to peers during classes and earn wallet rewards.

Priority Level	High	Complexity Rating	L
Execution Status	Backlog 📋	Target PR / Commit	`feature/P5-04-flex-share`
Upstream Dependencies	P5-03		
Architecture Blockers	None (Ready for execution)		

IMPLEMENTATION ROADMAP & ENGINEERING STEPS

1. Create FlexShareOffer and FlexShareClaim domain entities.
2. Implement slot matching service verifying return-time windows.
3. Implement idempotent reward credit service updating owner wallet upon release.
4. Create FlexShareController and abuse prevention guards.

TARGET FILE INVENTORY & EXPECTED ARTIFACTS

- `parking/domain/Flex*`
- `parking/application/Flex*`
- `parking/api/*`

ACCEPTANCE CRITERIA & QUALITY CHECKLIST

- Reward credited exactly once per verified exchange.
- Spot guaranteed available for primary permit holder upon scheduled return.

VERIFICATION & TEST COVERAGE STRATEGY

- FlexShareServiceTest
- ConcurrentClaimTest
- FlexShareControllerTest